Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese

Department of Electrical, Electronic and Computer Engineering

Semestertoets 2

Semester Test 2

Module EBN122

Elektrisiteit en Elektronika

October 2016

Module EBN122

Electricity and Electronics

October 2016

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Exam information: **Eksamensinligting:**			
Maximum marks: Maksimum punte:	45	*Full marks:* Volpunte:	46
Duration of paper: Duur van vraestel:	*90 minutes* 90 minute	*Open / closed book:* Oopboek / toeboek:	*Closed* Toe
Total numer of pages (including this page): Totale aantal bladsye (hierdie blad ingesluit):	13		

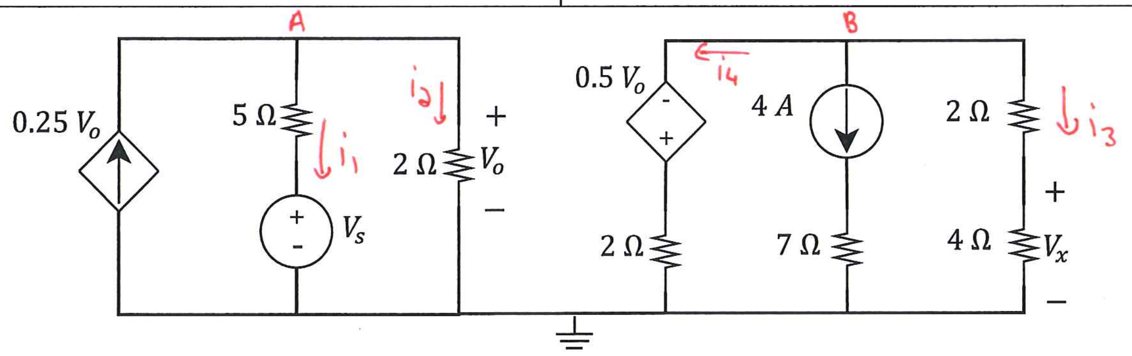
IMPORTANT **- BELANGRIK**

1. *The examination regulations of the University of Pretoria apply.*
   Die eksamenregulasies van die Universiteit van Pretoria geld.
2. *All questions must be answered on the ANSWER SHEET.*
   Alle vrae moet op die ANTWOORDBLAD beantwoord word.
3. *Students may do their own rough calculations on the question paper – the question paper will not be taken in.*
   Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
4. *Answers must include units!*
   Antwoorde moet eenhede insluit!
5. ***Final answers must be written as real numbers, and not as fractions.***
   Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.

Dosent(e): *Lecturer(s):*	1. D le Roux
	2. R dos Santos
	3. J Joubert

VRAAG/QUESTION 1 [8]

Question 1.1 [4]	**Vraag 1.1 [4]**
Using nodal analysis solve for V_s and V_x, if given that $V_o = 8\,V$.	Gebruik node analise om V_S en V_X op te los, indien gegee dat $V_0 = 8\,V$.



KCL at A: $0.25V_o = i_1 + i_2$

$$0.25V_o = \frac{V_o - V_s}{5} + \frac{V_o}{2}$$

$V_o = 8V$

$$2 = \frac{8 - V_s}{5} + 4$$

$$10 = 8 - V_s + 20$$

$$V_s = 18V$$

KCL at B: $i_4 + 4 + i_3 = 0$

$$\frac{V_B - (-0.5V_o)}{2} + 4 + \frac{V_B - 0}{2+4} = 0$$

$V_o = 8V$

$$\frac{V_B + 4}{2} + \frac{V_B}{6} = -4$$

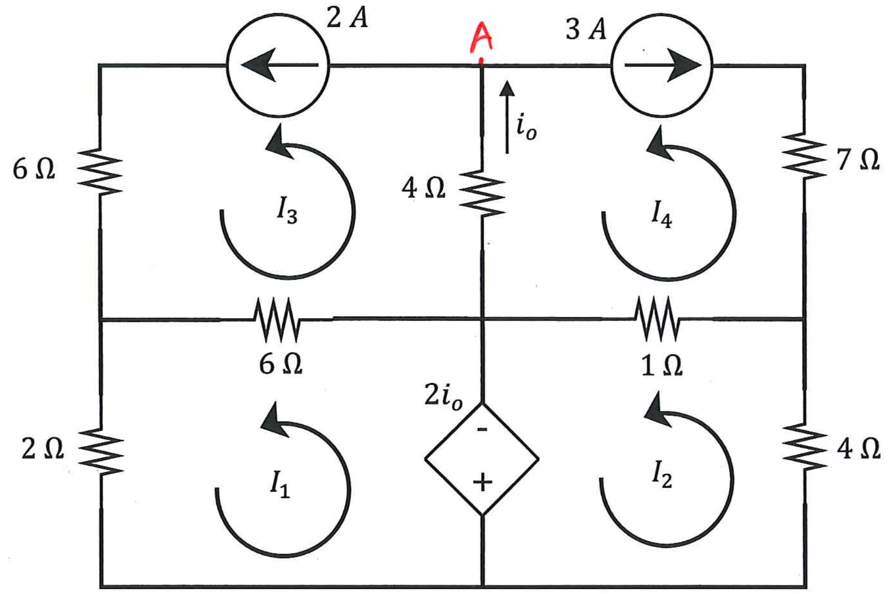
$$3V_B + 12 + V_B = \cancel{4} -24$$

$$4V_B = -36$$

$$V_B = -9V$$

Voltage Division: $V_x = \frac{4 \times V_B}{2+4} = \frac{4 \times -9}{6} = -6V$

Question 1.2 [4] Using mesh analysis determine I_1 and I_2.	**Vraag 1.2 [4]** Gebruik lusanalise om I_1 en I_2 op te los.



$I_3 = 2A$

$I_4 = -3A$

KCL A: $i_o = 2 + 3 = 5A$

Loop 1: $2I_1 + 6(I_1 - I_3) + 2i_o = 0$

$8I_1 - 6I_3 = -2i_o$

$8I_1 = 6I_3 - 2i_o$

$8I_1 = 6 \times 2 - 2 \times 5$

$I_1 = \frac{2}{8} = \underline{0.25A}$

Loop 2: $4I_2 + 1(I_2 - I_4) - 2i_o = 0$

$5I_2 - I_4 = 2i_o$

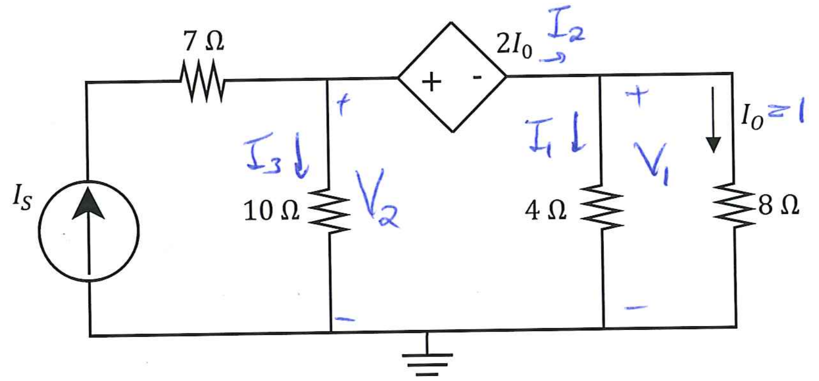
$5I_2 = 2i_o + I_4$

$5I_2 = 2 \times 5 + (-3)$

$I_2 = \frac{7}{5} = \underline{1.4A}$

VRAAG/QUESTION 2 [22]

Question 2.1 [2]	**Vraag 2.1 [2]**
Using the principle of linearity, determine the ratio: $\frac{I_0}{I_S}$.	Deur gebruik te maak van die beginsel van lineariteit, bepaal die verhouding: $\frac{I_0}{I_S}$.



* Assume $I_0 = 1A$.

→ $V_1 = 8V$

→ $I_1 = 8/4 = 2A$.

→ $I_2 = I_1 + I_0 = 3A$.

* KVL: $-V_2 + 2I_0 + V_1 = 0$

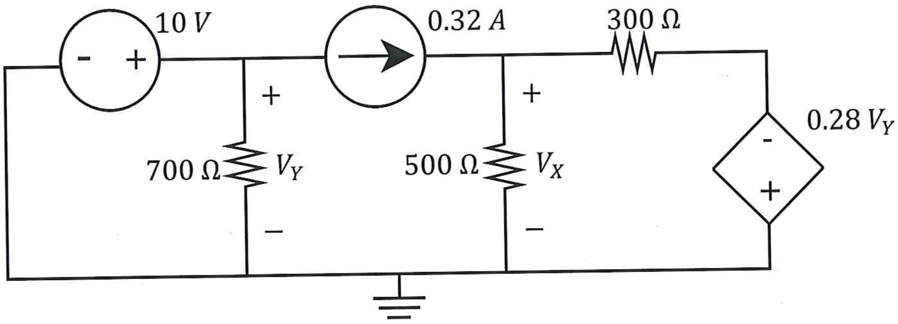
$V_2 = 2(1) + 8 = 10V$.

→ $I_3 = \frac{10}{10} = 1A$

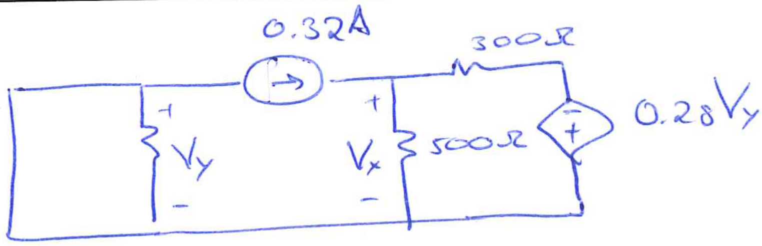
→ $I_S = I_2 + I_3 = 3 + 1 = 4$.

$$\frac{I_0}{I_S} = \frac{1}{4} = \underline{0.25}$$

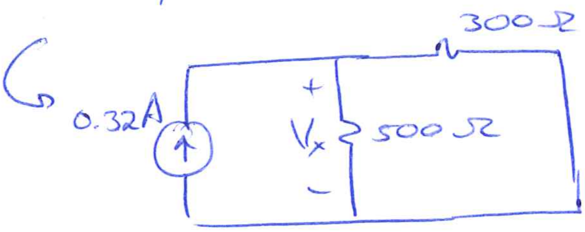
Question 2.2 [4]	Vraag 2.2 [4]
a) Determine the individual contribution of the 320 mA current source to V_X. b) Determine the individual contribution of the 10 V voltage source to V_X.	a) Bepaal die unieke bydra van die 320 mA stroombron tot V_X. b) Bepaal die unieke bydra van die 10 V spanningsbron tot V_X.



a)



$\rightarrow V_y = 0V$

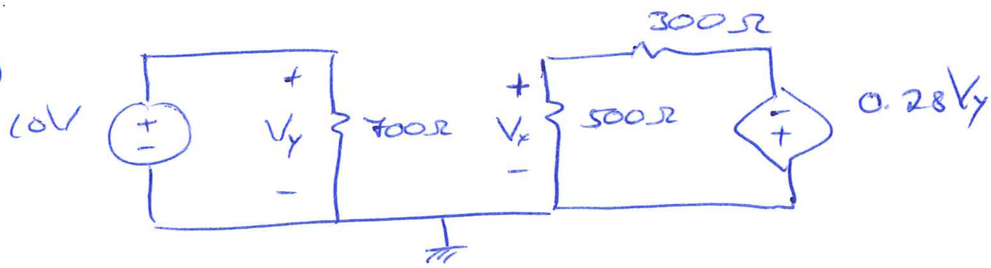


$$V_x = 0.32 \times (300 \parallel 500)$$

$$= \frac{0.32 \times 300 \times 500}{800}$$

$$= \underline{60V}$$

b)

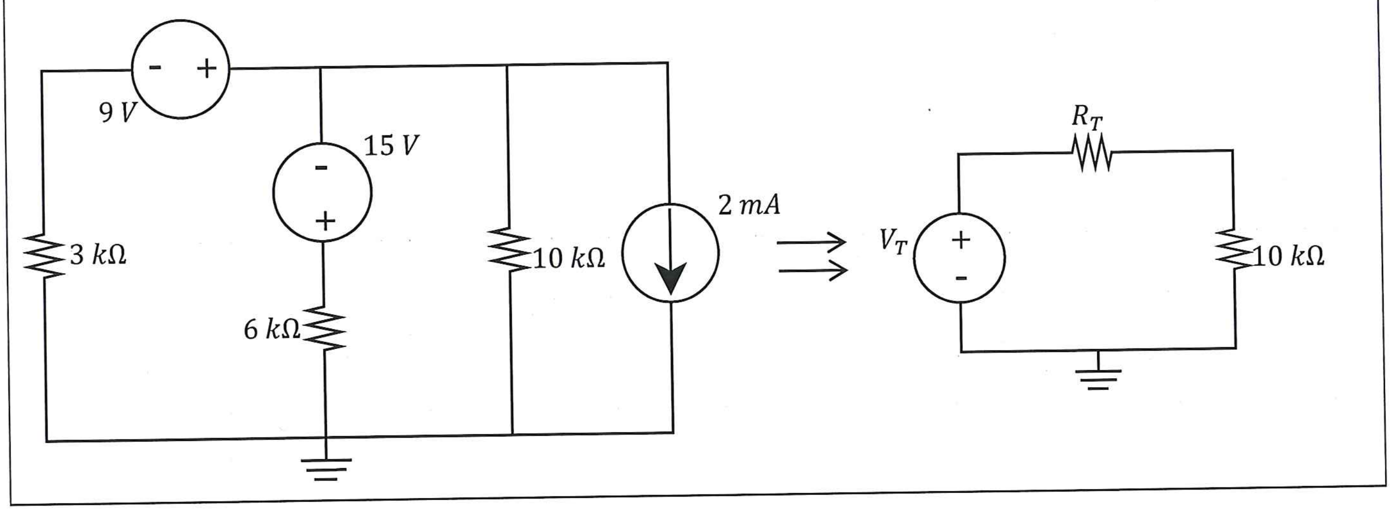


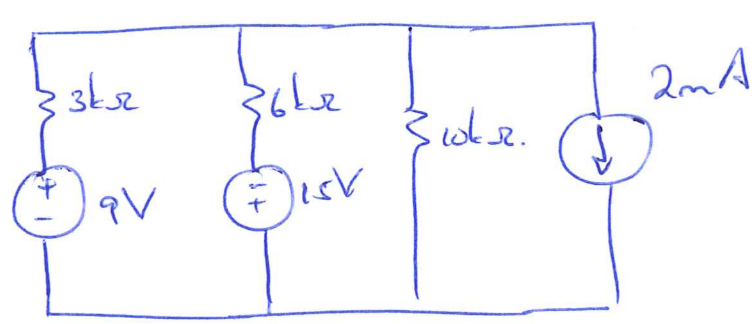
$\rightarrow V_y = 10V.$

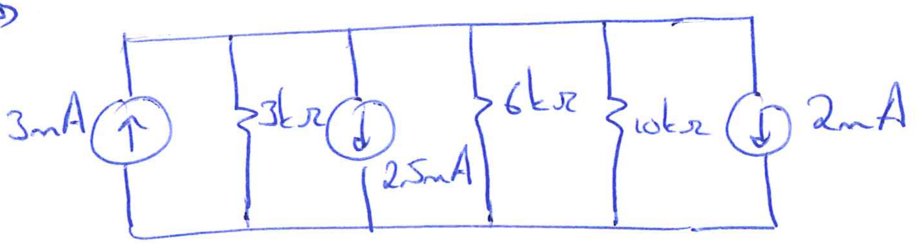
$\rightarrow V_x = \frac{500}{800} \times (-0.28V_y)$ (Voltage div).

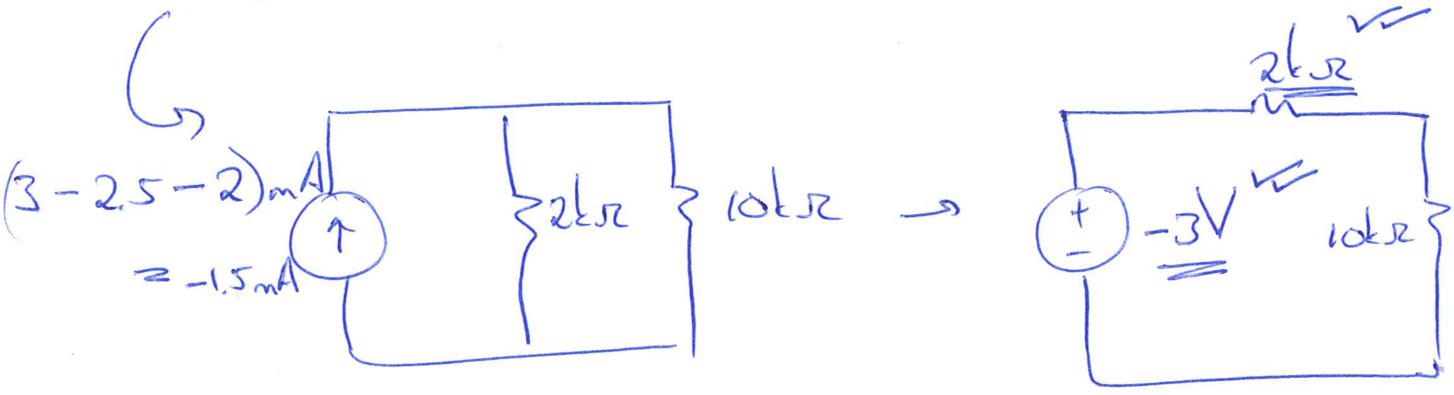
$= \underline{-1.75V}$

Question 2.3 [4] The aim is to simplify the circuit on the left to obtain the circuit on the right with respect to the 10 kΩ resistor. Make use of source transformation and determine V_T and R_T.	**Vraag 2.3 [4]** Die doel is om die linkerkantse kringbaan te vereenvoudig na die regterkantse kringbaan met verwysing tot die 10 kΩ weerstand. Maak gebruik van brontransformasie en bepaal V_T en R_T.



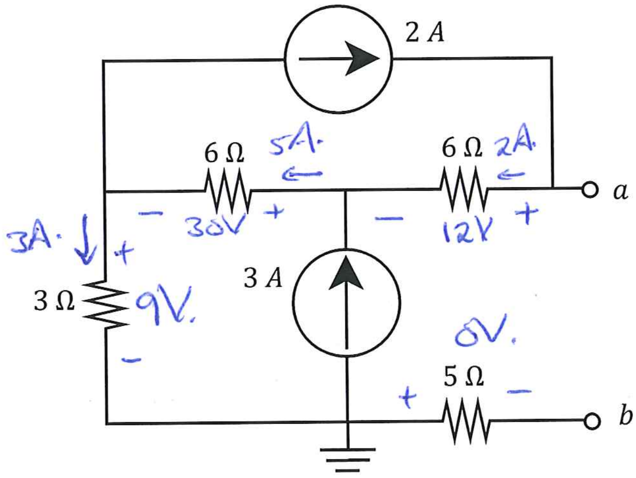






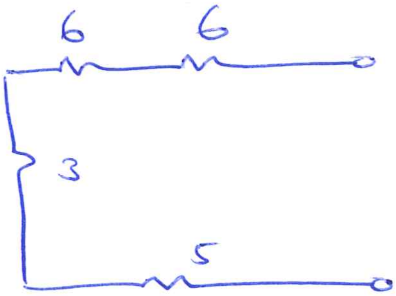
$$R_{ll} = 6k \| 3k = \frac{6\times 3}{6+3} = \underline{2k\Omega}$$

Question 2.4 [4]	Vraag 2.4 [4]
Determine the Thevenin equivalent voltage V_{TH} and resistance R_{TH} w.r.t. terminals *a-b*.	Bepaal die Thevenin ekwivalente spanning V_{TH} en weerstand R_{TH} t.o.v. terminale *a-b*.



* $\underline{V_{TH}}$ KVL: $V_{ab} = 12 + 30 + 9 = \underline{51V}$ ✓✓

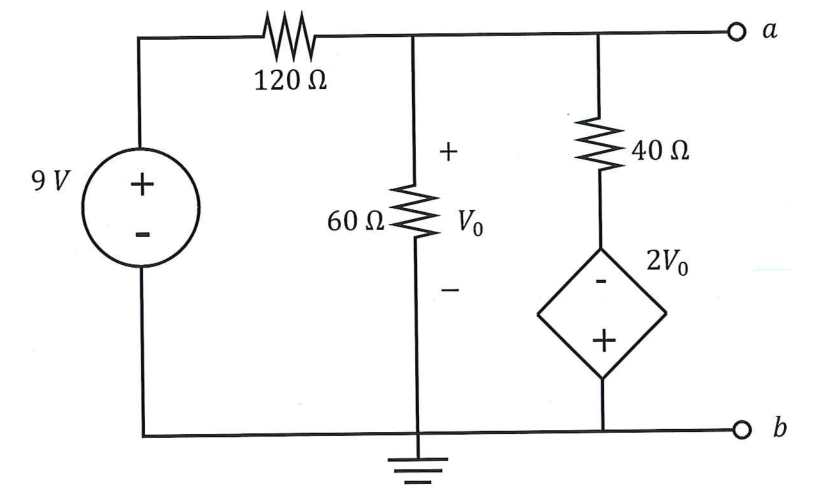
* $\underline{R_{TH}}$:



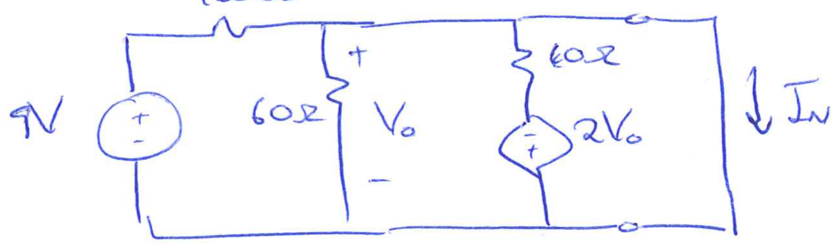
$R_{TH} = 6+6+3+5$

$= \underline{20\Omega}$ ✓✓

Question 2.5 [4]	Vraag 2.5 [4]
Determine the Norton equivalent current I_N and resistance R_N wrt terminals *a-b*.	Bepaal die Norton ekwivalente stroom I_N en weerstand R_N tov terminale *a-b*.

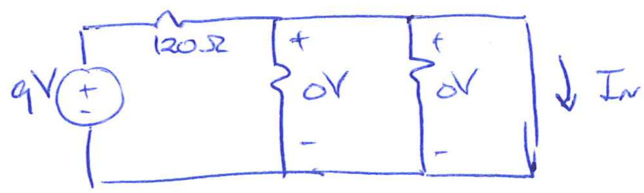


$\underline{\underline{I_N}}$

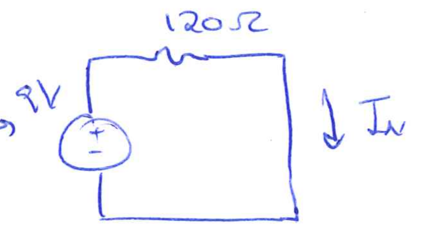


↳ $V_o = 0V.$

↳

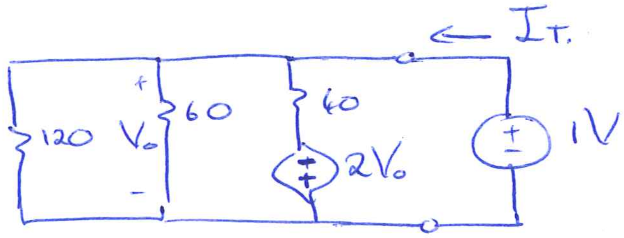


→



$$I_N = \frac{9}{120} = 75mA$$ ✓✓

$\underline{\underline{R_N}}$

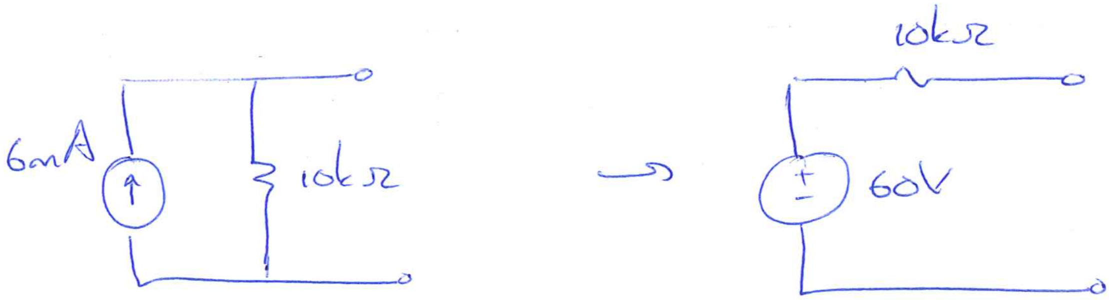


↳ $V_o = 1V.$

$$* \quad I_T = \frac{1+2}{40} + \frac{1}{60} + \frac{1}{120} = \frac{1}{10}$$

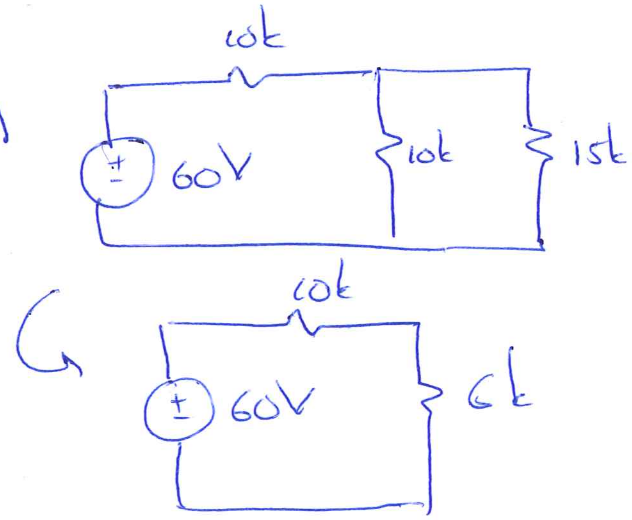
$$R_N = \frac{1}{I_T} = 10\Omega$$ ✓✓

Question 2.6 [4] The short circuit current of a circuit is $I_N = 6$ mA. The circuit delivers maximum power to a load of $R_L = 10$ kΩ. a) What is the maximum power P_{max} absorbed by the load resistor R_L? b) If a resistor of 15 kΩ is placed in parallel to the load resistor R_L, what is the total power absorbed by the parallel combination? (I.e. the total power absorbed by the combined resistance (15 ‖ 10) kΩ.)	**Vraag 2.6 [4]** Die kortsluitstroom van `n kringbaan is $I_N = 6$ mA. Die kringbaan lewer maksimum drywing aan 'n las van $R_L = 10$ kΩ. a) Wat is die maksimum drywing P_{max} wat deur die lasweerstand verkwis word? b) Indien 'n weerstand van 15 kΩ parallel aan die las weerstand R_L gekoppel word, wat is die totale drywing wat deur die parallele kombinasie geabsorbeer word? (M.a.w. die total drywing geabsorbeer deur die gekombineerde weerstand (15 ‖ 10) kΩ.)



a) $P_{max} = \frac{V_{TH}^2}{4 R_{TH}} = \frac{60^2}{4 \times 10k} = \underline{90 \text{ mW}}$

b)

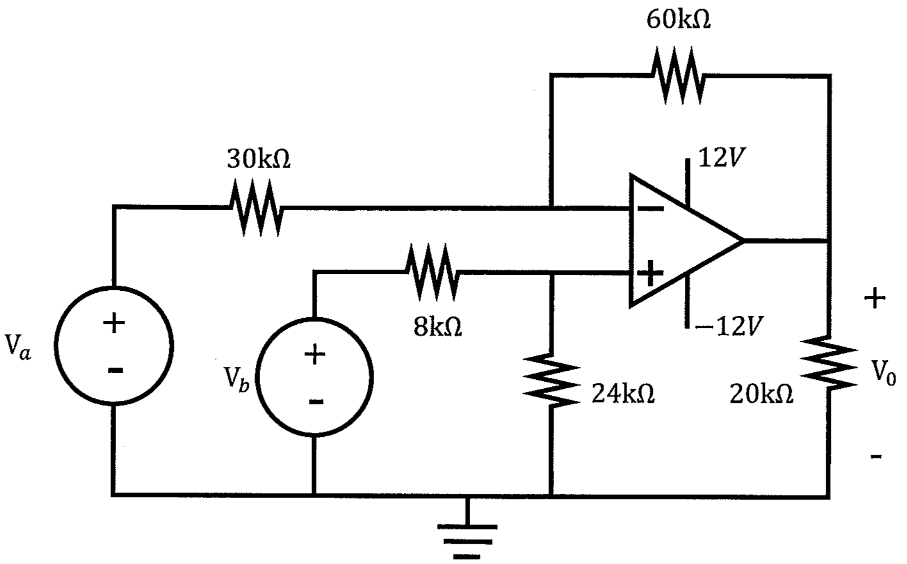


$P_6 = i^2 \cdot R = \left(\frac{60}{16k}\right)^2 \times 6k$

$= \underline{84{,}375 \text{ mW}}$

VRAAG/QUESTION 3 [16]

Question 3.1 [4]	**Vraag 3.1 [4]**
a) Calculate V_0 if $V_a = 2V$ and $V_b = 1V$. b) Calculate V_0 if $V_a = -3V$ and $V_b = 0V$.	a) Bereken V_0 as $V_a = 2V$ en $V_b = 1V$. b) Bereken V_0 as $V_a = -3V$ en $V_b = 0V$.



a) $V_b = 1V \Rightarrow V^+ = \frac{24}{32} \times 1 = 0.75V$

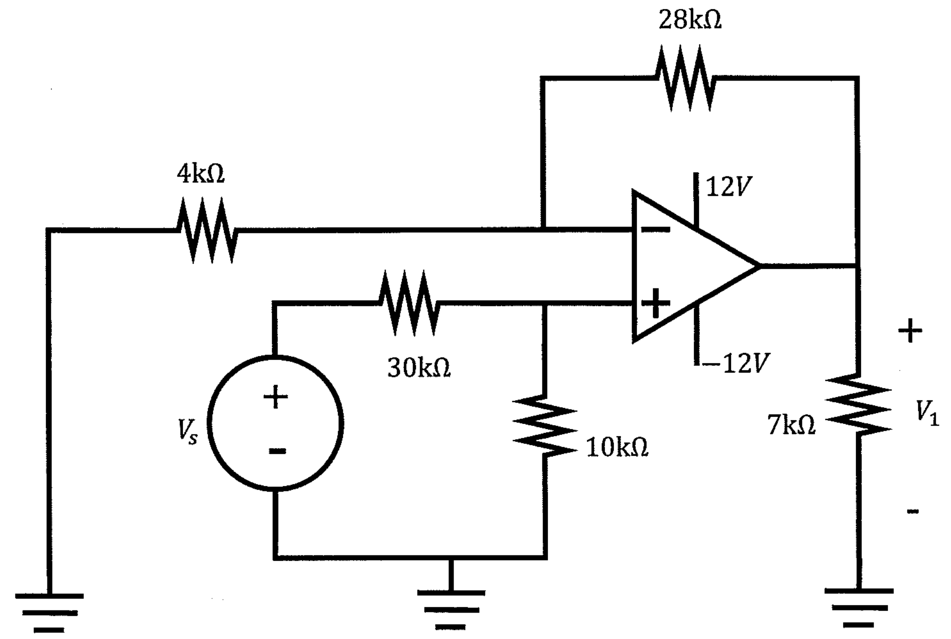
KCL @ V^-: $\frac{0.75-2}{30k} + \frac{0.75-V_0}{60k} = 0$

$\times 60k$: $2(-1.25) + 0.75 - V_0 = 0$

$\Rightarrow V_0 = -1.75V$ ✓✓

b) $V_0 = -\frac{60}{30} \times (-3) = 6V$ ✓✓

Question 3.2 [4]	Vraag 3.2 [4]
a) Determine V_1 if $V_s = 4V$. b) Determine the range of values of V_s that will ensure that the op-amp operates in the linear region.	a) Bepaal V_1 as $V_s = 4V$. b) Bepaal die omvang van waardes van V_s wat sal verseker dat die operasionele versterker in sy lineêre gebied sal funksioneer.



a) $V_s = 4 \Rightarrow V^+ = \frac{10}{40} \times 4 = 1\,V$

KCL at V^-: $\frac{1-0}{4k} + \frac{1-V_1}{28k} = 0$

$\times 28k$: $7 + 1 - V_1 = 0$

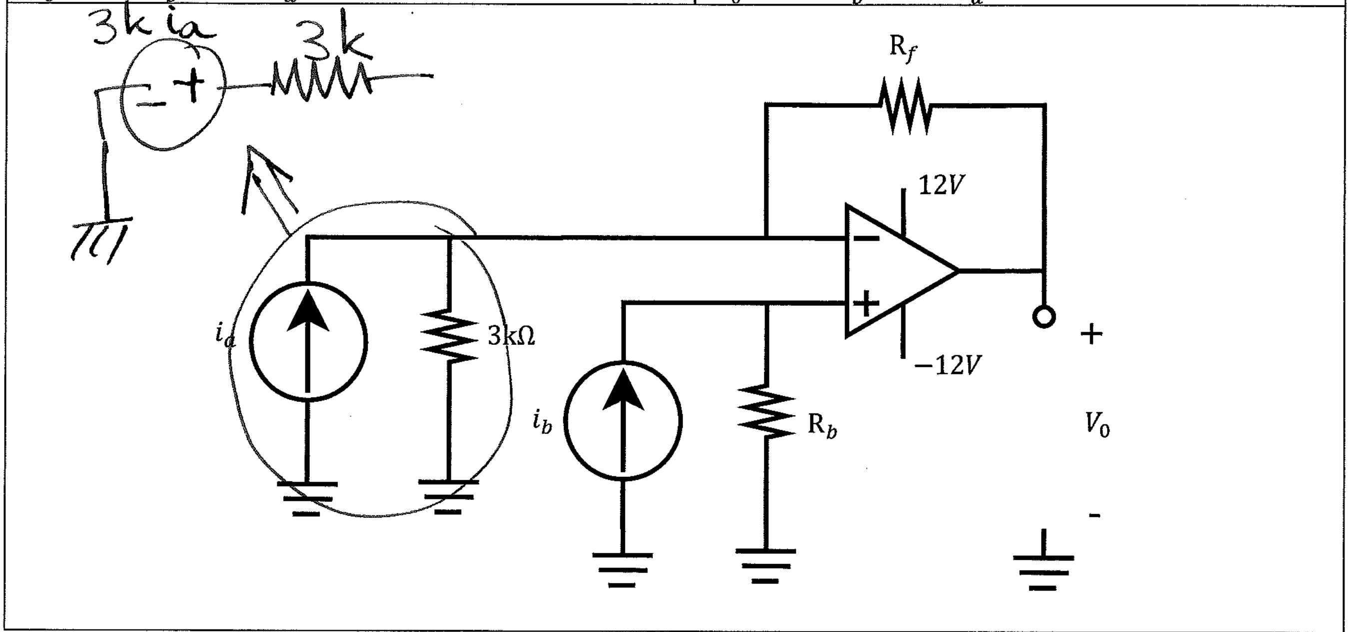
$\Rightarrow V_1 = 8V$

b) $\frac{\frac{V_s}{4} - 0}{4k} + \frac{\frac{V_s}{4} - V_1}{28k} = 0 \Rightarrow V_1 = 2V_s$

$\Rightarrow -6 < V_s < 6$

will imply $-12 < V_1 < 12$

Question 3.3 [4]	Vraag 3.3 [4]
Determine values of R_b and R_f so that $V_0 = 2000i_b - 2000i_a$.	Bepaal waardes vir R_b en R_f sodat $V_0 = 2000i_b - 2000i_a$.



Superposition:

$$V_0 = i_b R_b \times \left(\frac{R_f}{3k} + 1\right) \quad \text{(non inverting)}$$

$$- \cancel{3k}\, i_a \cdot \frac{R_f}{\cancel{3k}} \quad \text{(inverting)}$$

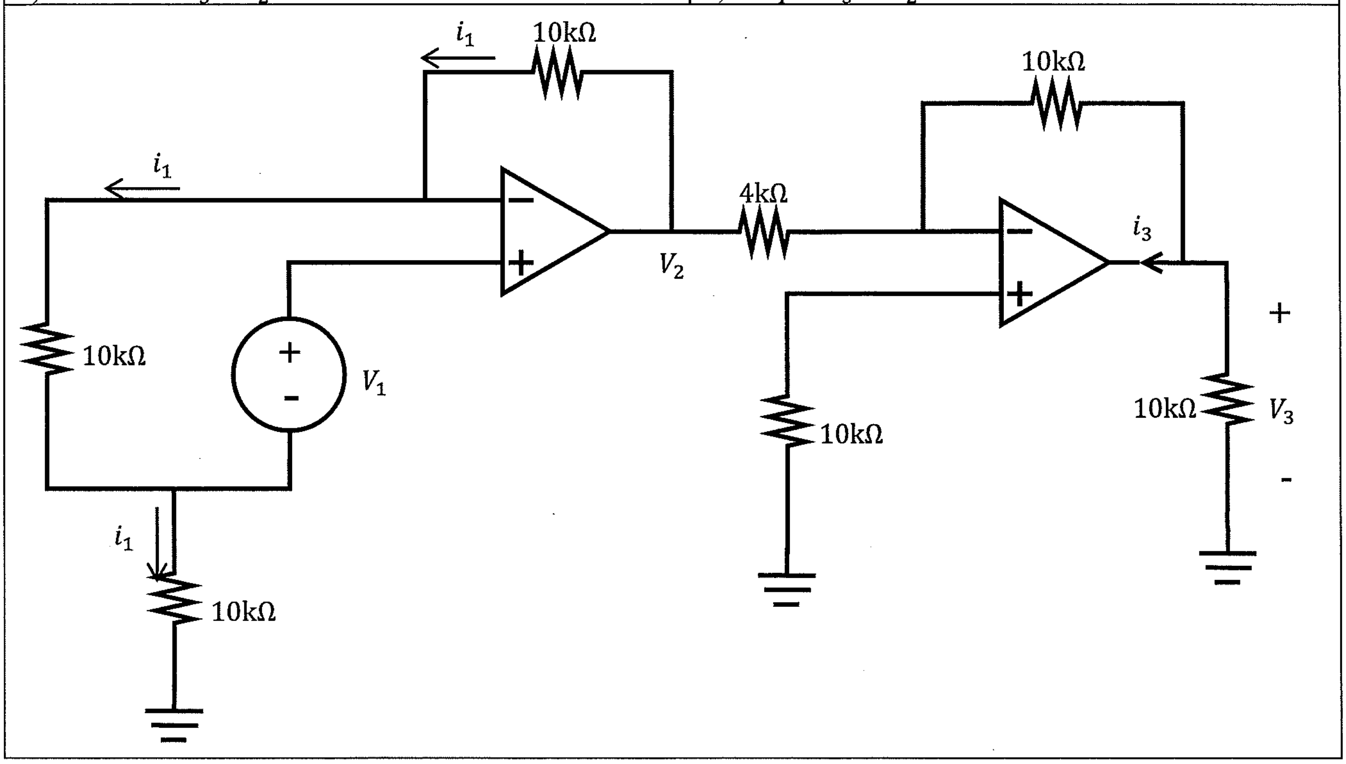
$$= 2000 i_b - 2000 i_a$$

$$\Rightarrow R_f = 2000\,\Omega$$

$$R_b\left(\frac{R_f}{3k} + 1\right) = 2000 \Rightarrow R_b = \frac{2000}{\frac{2000}{3000} + 1}$$

$$= 1200\,\Omega$$

Question 3.4 [4]	Vraag 3.4 [4]
a) Determine the amplification factor V_2/V_1.	a) Bepaal die versterkingsfaktor V_2/V_1.
b) Determine i_3 if $V_2 = 2V$.	b) Bepaal i_3 as $V_2 = 2V$.



a) $V^+ = i_1 \times 10k + V_1$

$V^- = i_1 \times 10k + i_1 \times 10k = V^+$

$\Rightarrow V_1 = i_1 \times 10k$

$V_2 = 3 \times i_1 \times 10k = 3V_1$

$\Rightarrow V_2/V_1 = 3$ ✓✓

b) $V_3 = -2 \times \frac{10}{4} = -5V$

KCL: $i_3 + \frac{(-5)}{10k} + \frac{(-5)}{10k} = 0$

$\Rightarrow i_3 = 1mA$ ✓✓